Abhishek Yadav

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Summary

Backend engineer with hands-on experience in data pipelines, microservices, and full-stack development using Java, Python, FastAPI, and Spring Boot. Passionate about building scalable systems with cloud-native architectures, API design, and automation. Strong foundation in REST APIs, CI/CD pipelines, unit testing, and AI-integrated applications. Focused on backend optimization and data quality in production environments.

Work Experience

Software Development Engineer (SDE) at Anantasutra May 2026 – Present

- Built scalable Python-based web scraping and lead-generation pipelines using **BeautifulSoup, Requests, REST APIs, and microservices**, increasing data acquisition by **40%** while improving data quality by **35%** through automated verification, deduplication, and backend optimization.
- Developed **GitHub Actions CI/CD** pipelines with automated testing and deployment, implemented unit testing and backend optimizations, reducing invalid records by **30%** and improving the reliability, maintainability, and scalability of data automation systems.

Data Engineer Intern at Anantasutra Nov 2025 – Jan 2026

- Built automated lead-generation pipelines via web scraping with Python (BeautifulSoup, Requests) and RESTful API integrations, increasing data acquisition by 40% while improving data quality by 35% through automated verification, deduplication, and backend optimization.
- Developed microservices-based automation with GitHub Actions CI/CD and unit testing, reducing invalid records by 30%, automating deployment and testing, and improving system reliability and scalability.

Projects

Distributed Key-Value Store GitHub Repo

Built a production-grade in-memory KV store from scratch in Java 17: TCP server with FixedThreadPool (50 concurrent clients), LRU eviction via LinkedHashMap (O(1) all ops), dual-strategy TTL expiry (lazy deletion + ScheduledExecutorService sweep), and AOF write-ahead log with fsync policies and atomic compaction for crash recovery. 144 unit and integration tests (JUnit 5) including concurrency and full durability lifecycle tests. Deployable as a standalone JAR with an CLI client.

Voice AI Platform — Full-Stack Voice AI SaaS GitHub Repo

Built a cloud-native Voice AI SaaS platform with microservices backend (FastAPI, MongoDB, Redis, Celery, Docker), JWT auth, WebSocket metrics, LiveKit SIP + Vobiz telephony, OpenAI/Gemini voice models, Deepgram STT, and Sarvam TTS with multi-language support. CI/CD pipelines via GitHub Actions. Frontend with React (TypeScript), Vite, Tailwind CSS, Zustand, and RBAC-based admin dashboard for user management, billing, and real-time analytics.

Nexus Control Center GitHub Repo

Developing a modern Linux command center using Rust, Tauri v2, React, and TypeScript to provide a unified desktop application for system management, performance monitoring, and hardware control. Designed a modular plugin-based architecture with real-time CPU, memory, disk, battery, and network monitoring, enabling scalable feature development and efficient system resource management. Integrating AI-powered diagnostics, automation workflows, and persistent configuration management following modern software architecture and desktop application design principles.

Education

Bachelor of Technology in Computer Science and Engineering 2023 – 2027 (Expected)

Veer Bahadur Singh Purvanchal University

Skills

Languages: Java, Python
Backend: Spring Boot, FastAPI, REST APIs, Microservices
Cloud & DevOps: Docker, CI/CD, Cloud-Native
API & Auth: API Design, WebSocket, JWT, RBAC
Databases: MongoDB, MySQL, Redis, Celery
Testing & Quality: Unit Testing, Data Quality
AI/ML: OpenAI, Gemini, Deepgram, Sarvam
Data: Data Pipelines, Web Scraping, Automation
Frontend: React, TypeScript, HTML/CSS
Tools: Postman, Git, VS Code, Cursor, IntelliJ IDEA

Additional Information & Achievements

- **Certificates:** Deloitte Australia Technology Job Simulation on Forage - July 2025 | CodSoft – Java Programming | Bharat Innovation Conclave – Participation | Cognifyz Technologies – Java Development